VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2008

H2 CHEMISTRY 9746/3

PAPER 3

Monday 2 hours
08/09/2008 80 marks

Additional materials:
- Answer Paper
- Graph Paper
- Data Booklet

INSTRUCTIONS TO CANDIDATES

Write your *name* and *CT group* in the spaces provided on the answer paper.

Answer any **four** questions.

Begin each question on a **fresh sheet** of paper.

Write the **question numbers** of **all** the questions you have attempted on the **first** answer script.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You may use a calculator.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

1

(a) Calcium hydroxide is sparingly soluble in water. The numerical value for the K_{sp} of $Ca(OH)_2$ is 2.2×10^{-5} at 25°C.

(i) Write an expression for the K_{sp} of calcium hydroxide.
Hence, calculate $[Ca^{2+}(aq)]$ in a saturated solution of calcium hydroxide.

In an experiment to determine the K_{sp} value of calcium hydroxide, an excess of solid calcium hydroxide is shaken in 0.100 mol dm^{-3} sodium hydroxide to reach equilibrium at 25°C and then filtered. 10.0 cm^3 of the filtered solution is required to neutralise 10.40 cm^3 of 0.100 mol dm^{-3} hydrochloric acid.

(ii) Calculate the concentration of calcium ions, in mol dm^{-3}, in the solution.

(iii) Hence, calculate the solubility product of calcium hydroxide, stating the units.

(iv) Comment on your answers for **(i)**, **(ii)** and **(iii)**.

[8]

(b) Calcium reacts with water to form aqueous calcium hydroxide.

$$Ca(s) + 2H_2O(l) \xrightarrow{\Delta H} Ca^{2+}(aq) + 2OH^{-}(aq) + H_2(g)$$

Some enthalpy changes are listed below.

Enthalpy change of atomisation of calcium	=	178 kJ mol^{-1}
Enthalpy change of neutralisation	=	–58 kJ mol^{-1}
Enthalpy change of hydration of $Ca^{2+}(g)$	=	–1577 kJ mol^{-1}
Enthalpy change for $2H^{+}(aq) \rightarrow H_2(g)$	=	–796 kJ mol^{-1}

By using data in this question and relevant data in the *Data Booklet*, draw an energy level diagram to determine ΔH for the above reaction. [5]

(c) The lattice energy of a metal is small compared with that of an ionic compound. The lattice energy of calcium is –178 kJ mol^{-1} and that of calcium hydroxide is –2506 kJ mol^{-1}.

Describe the structure and bonding of Ca and $Ca(OH)_2$. Hence, comment on the difference between their lattice energies. [4]

(d) Both calcium and beryllium are Group II elements. However, their compounds differ greatly in their boiling points.

Compound	Boiling point / °C
$BeCl_2$	482
$CaCl_2$	1940

Explain the difference in their boiling points. [3]

[Total: 20]

2

(a) The kinetics of the hydrolysis of the ester, RCO_2CH_3, may be investigated by determining the concentration of RCO_2H produced.

$$RCO_2CH_3 + H_2O \rightarrow RCO_2H + CH_3OH$$

In a 1 dm^3 mixture, 0.350 mol of the ester was hydrolysed by heating with water and using hydrochloric acid as catalyst. The following results were obtained.

Time / s	Concentration of RCO_2H / mol dm^{-3}
0	0
340	0.105
680	0.185
1080	0.243
1440	0.278

(i) By drawing a suitable graph using the data given above, show that the reaction is first order with respect to the ester.

It has been found that the hydrolysis reaction is first order with respect to the hydrochloric acid.

(ii) Write down the rate equation for this reaction and state the units of the rate constant.

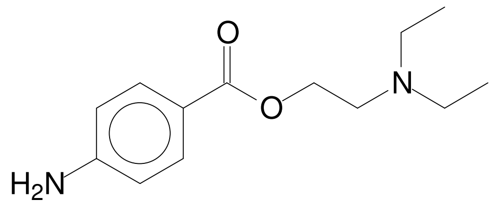
(iii) State how the half-life of the reaction is affected when the experiment is repeated with the following changes.

(I) Concentration of HC*l* is doubled.

(II) Concentration of RCO_2CH_3 is doubled.

[8]

(b) Procaine, a first injectable local anesthetic used during surgery, is an ester with the following structure.



Procaine

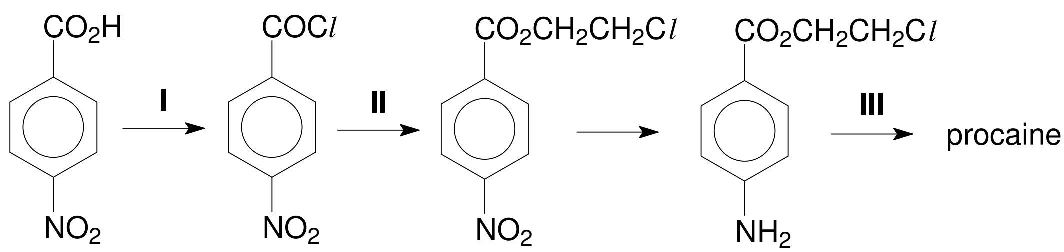
(i) One molecule of procaine contains two nitrogen atoms, both of which can act as a base by accepting a proton. Copy the structure of procaine and circle the nitrogen atom which you think is the stronger base. Explain your reasoning.

(ii) Hence deduce the structural formula of the product formed when one mole of procaine is treated with **one** mole of HC*l* at room temperature.

(iii) Procaine undergoes hydrolysis rapidly in the small intestines where the pH is about 8. Draw the structural formulae of the products obtained from the hydrolysis.

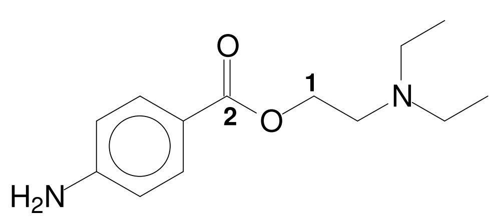
(iv) Procaine has a half–life of about 40 to 84 seconds and is rarely used today since more effective alternatives such as lidocaine exist. Lidocaine has a half–life of about 1.5 to 2 hours. Give a reason why lidocaine is more effective than procaine.

(v) Procaine can be made by the following reaction scheme:



State the reagents and conditions used for stages **I**, **II** and **III**.

(vi) The geometry (shape) with respect to carbon atoms labeled **1** and **2** in procaine are different.



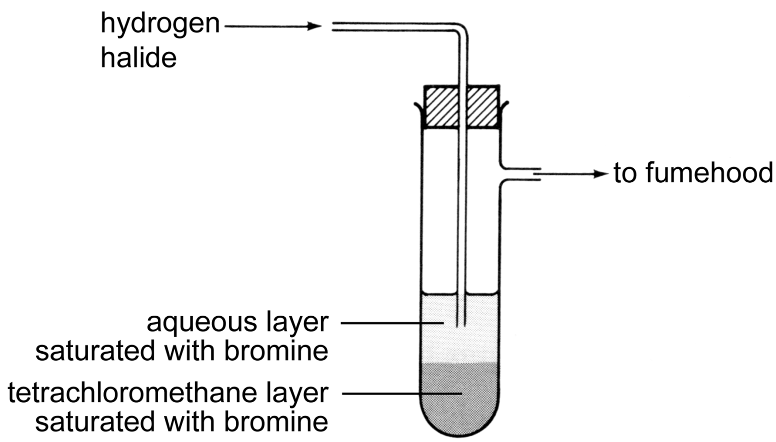
State the shape and type of hybrid orbitals at carbon atoms labeled **1** and **2**. Hence sketch the shapes of the hybrid orbitals around the two carbon atoms.

[12]

[Total: 20]

3

(a) A student devised the following set of apparatus to detect hydrogen halides.



(i) Suggest a hydrogen halide that can be detected by this system, giving the expected observations and equation.

(ii) In fact, this set–up does not work as expected. Why do you think the set–up cannot work?

[4]

(b) Chlorine (0.12 mol) was made to react with NaOH at a temperature that allowed **both** of the following reactions to occur.

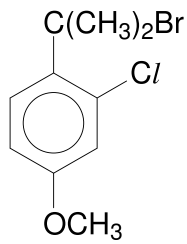
$$Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$$

$$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$$

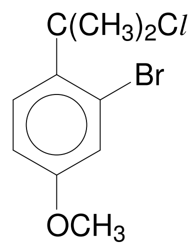
Immediately after all of the chlorine had reacted, it was found that the solution contained 0.16 mol of NaC*l*.

Calculate the number of moles of $NaClO_3$ produced. [2]

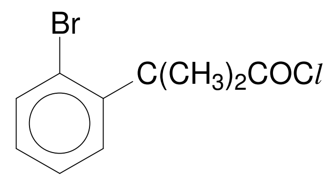
(c) Halogens are commonly found in organic compounds such as those shown below.



A



B

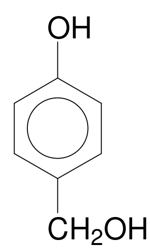


C

Explain using chemical tests how members of each of the following compounds can be distinguished from one another. Make clear the observations you would expect to make with each member. Write equations for the reactions that occur for **A**.

[The group $-OCH_3$ which is present in **A** and **B** can be regarded as inert.] [4]

(d) When an organic halogen compound **W**, $C_{10}H_9O_3Br$, is boiled with aqueous sodium hydroxide followed by acidification, it gives two compounds, one of which has the following structure shown below.



When sodium hydrogencarbonate is added to the other compound **X**, effervescence of carbon dioxide is observed. In addition, **X** gives a yellow precipitate on warming with alkaline aqueous iodine. When **X** is treated with an ethanolic solution of $NaBH_4$, it gives **Y**. Heating **Y** in the presence of an acid catalyst gives a sweet-smelling liquid, **Z**, $C_6H_8O_4$.

Deduce the structures for each lettered compound, **W** to **Z**, and give an account of the chemistry involved. [10]

[Total: 20]

4

(a) The Edison or nickel–iron battery is a rechargeable battery in which nickel and iron electrodes, coated with their respective hydroxides, are immersed in an electrolyte consisting of potassium hydroxide.

The relevant electrode reactions are

$NiO(OH)(s) + H_2O(l) + e^- \rightleftharpoons Ni(OH)_2(s) + OH^-(aq)$ $\quad E^\ominus = +0.48V$

$Fe(OH)_2(s) + 2e^- \rightleftharpoons Fe(s) + 2OH^-(aq)$ $\quad E^\ominus = –0.90V$

(i) Draw a diagram for this cell, showing the polarity of the electrodes during discharge.

(ii) Calculate the value of $E^\ominus_{cell}$ for this cell reaction. Write a balanced equation for the reaction that occurs during discharge.

[4]

(b) When 1.00 g of a red solid **D**, with formula K_xNiF_6, was reacted with water, 48 cm^3 of oxygen (measured at 298 K and 1 atm) was evolved and a green acidic solution, **E**, was formed. **E** consists of KF, NiF_2 and HF.

Solution **E** was divided into 2 equal parts.

Titration of one part with 0.20 mol dm^{-3} NaOH required 19.90 cm^3 for neutralisation.

The other part was electrolysed using a current of 0.40 A and took 16 minutes to completely deposit the nickel metal at the cathode.

(i) Calculate the number of moles of oxygen evolved.

(ii) Calculate the number of moles of hydrogen ions and nickel ions in solution **E**.

(iii) Using your answers to **(i)** and **(ii)**, deduce the value of *x* with the aid of an equation.

(iv) Explain why the complex ion, NiF_6^{x-}, found in solid **D** is coloured.

[8]

(c) Transition elements such as cobalt also forms coloured complexes.

(i) State the coordination number in $[Co(en)_3]^{3+}$ where en refers to ethylenediamine, $H_2NCH_2CH_2NH_2$.

(ii) The standard electrode potential of the $[Co(en)_3]^{3+}$ ion is given below.

$$[Co(en)_3]^{3+}(aq) + e^- \rightleftharpoons [Co(en)_3]^{2+}(aq) \qquad E^\ominus = +0.18V$$

By means of a fully labeled diagram, describe how the standard electrode potential of the $[Co(en)_3]^{3+} | [Co(en)_3]^{2+}$ system can be measured.

(iii) The presence of ethylenediamine significantly decreases the oxidising ability of Co(III). Illustrate this effect, selecting appropriate $E^\ominus$ values from the *Data Booklet*.

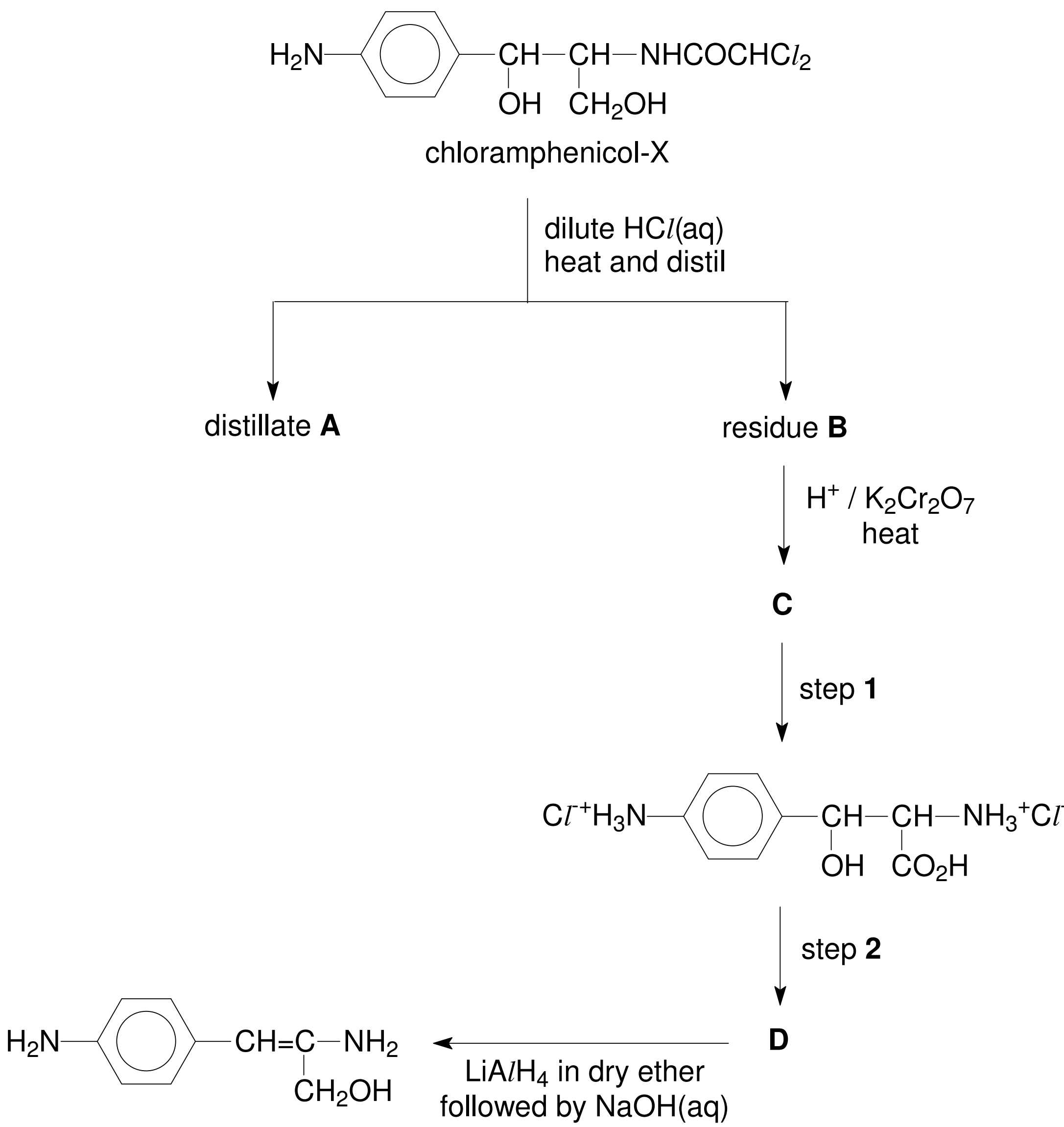
(iv) Bubbling of oxygen gas through a strongly acidic pink solution of $CoSO_4$ does not give any observable changes. However, the use of fluorine gas leads to the formation of a clear blue solution. When left to stand, the blue solution gradually turns pink with the evolution of a colourless gas.

Use the information given above, and data from the *Data Booklet*, suggest explanations for the observations.

[8]

[Total: 20]

5 Chloramphenicol-X, an antibiotic drug, can be converted into compound **E** via a series of reactions shown below.



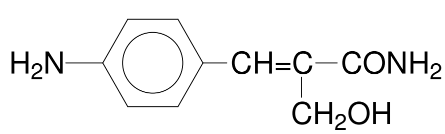
E

(a) Identify the lettered compounds **A** – **D** in the above scheme of reactions. [4]

(b) State the reagents and conditions used for steps **1** and **2**. [2]

(c) State and explain whether you would expect chloramphenicol–X to exist as a solid at room temperature. [2]

(d) Describe a simple chemical test to distinguish between **E** and **F**. [2]



F

(e) Draw the displayed formula of **E**, showing clearly any lone pair electrons, and state the following bond angles in your formula: H–N–H, N–C–C and C–O–H. [3]

(f) **E** can be converted into an amino acid. Amino acids can be linked together to form proteins. Draw the displayed formula of the linkage that joins the amino acids in proteins. [1]

(g) Haemoglobin is a protein with a quartenary structure. It is responsible for transporting oxygen from the lungs to all parts of the body.

(i) Explain what is meant by *quartenary structure*, specifying the protein components in haemoglobin.

(ii) Do you expect haemoglobin to be soluble in water? Explain your reasoning.

(iii) Name one poison that can affect haemoglobin, and describe how it reacts. [6]

[Total: 20]